From Loss to Lookup: Tracing Circuit Formation in a Small Transformer

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Abstract

Mechanistic interpretability usually studies finished circuits; this paper studies how a circuit forms during training. I use a controlled symbolic latest-write key-value lookup task, where the model must identify the queried key, retrieve the latest matching support value, and write value identity into the prediction-position residual stream. I define role-level progress measures for retrieval and write/readout coupling. In the traced reference run, the QK side forms as a low-rank W_{QK} support-value matcher, and first-order attribution using the actual AdamW parameter update tracks QK route growth while the instantaneous raw-gradient/SGD-equivalent direction explains little of the measured movement. Across additional seeds, a similar support-value retrieval role recurs while its implementing head changes. The write/readout side is causal but different: it is best characterized as a contextual, high-rank prediction-position value-code operation rather than a clean static W_{OV} theorem. The contribution is not a new primitive method, but a controlled formation audit that follows one retrieval role from behavior into route geometry, causal subspaces, optimizer-update attribution, cross-seed address movement, and write/readout structure.

1. Introduction

Mechanistic interpretability usually starts after training, using circuit and path-level analyses to explain already-trained behavior (Elhage et al., 2021; Wang et al., 2023; Goldowsky-Dill et al., 2023). A model behaves in some way; I inspect the trained weights and activations; I try to identify the circuit that implements the behavior. This paper asks how training writes a circuit into the model in the first place. The distinction matters because a finished circuit and a forming circuit need not have the same natural unit of explanation.

The central problem is that component names can be unstable. Following the progress-measure style of tracking learned algorithmic structure over training (Nanda et al., 2023), I use "role" to mean a task-level computational function measured by a scalar functional, independent of which component implements it. The retrieval role is measured by C_{QK}: support-value score above distractors. The write role is measured by C_{write}: a prediction-position residual change aligned with a mature answer-relevant direction. This is a measurement choice, not a new ontology: it is useful here because the implementation is dense and partially superposed.

Within this setup, the measured role is more stable than the component address. Across random seeds, the same role reappears, but the winning head or path moves.

I use a small symbolic task because it makes this question measurable. The model is a 3-layer decoder-only transformer trained on latest-write key-value lookup. The task is simple enough to audit, but rich enough to require routing, writing, and readout. I track the circuit from behavior to residual states, from residual states to weight geometry, from weight geometry to optimizer updates, and then across seeds.

The paper therefore makes a bounded claim. In this controlled symbolic transformer, the more stable unit of formation is a lookup role rather than a component address. Across seeds, the support-value retrieval role repeatedly appears while its implementing head changes. In the reference seed, the route side forms as a low-rank QK support-value matcher, and first-order attribution using the actual AdamW parameter update tracks its measured growth. The write side is causal but different: it is better described as a contextual, high-rank prediction-position value-code operation than as a static OV theorem. The audit is strong for QK, supported for write/readout subspaces, and deliberately bounded on optimizer generality and write-side closure.

Claims and evidence. I make three claims.

1. **Role/address dissociation.** The support-value retrieval role repeats across seeds, while the winning head address changes.

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2. **QK route formation.** In the reference seed, the route side forms as low-rank W_{QK} crystallization, and first-order attribution using actual AdamW parameter updates tracks the measured growth.

3. **Contextual write/readout.** The write side is causal and reproducible at the value-code/subspace level, but it is not reduced to a clean static W_{OV} theorem.

Compact proof ledger

claim	status
QK route	strong
QK birth	strong
AdamW update accounting	strong in traced run
role/address dissociation	strong
write/readout value code	causal/supported
static OV theorem	no
moving-margin closure	partial

2. Related Work and Positioning

This analysis builds on the transformer architecture (Vaswani et al., 2017) and on the transformer-circuits framing that separates attention into QK routing and OV writing (Elhage et al., 2021). It also follows the broader circuit-analysis tradition of explaining behavior through causal components and paths, including IOI-style circuit analysis (Wang et al., 2023), path patching (Goldowsky-Dill et al., 2023), and automated circuit discovery (Conmy et al., 2023). Those works primarily localize or validate mechanisms in trained models; my target is the formation trajectory of a measured role.

The closest precedent for studying circuit formation is induction-head formation (Olsson et al., 2022; Singh et al., 2024), which connects learned match-and-copy circuits to training-time phase changes and identifies subcircuits that must interact for induction heads to emerge. My setting studies a different retrieval problem—latest-write symbolic lookup—and focuses on optimizer-update accounting and a write/readout side that does not reduce to a clean static OV map.

The superposition framing follows (Elhage et al., 2022; Bricken et al., 2023). This is why the unit of explanation is a role/subspace rather than a neuron-level theorem. Distributed alignment work also argues that interpretable causal variables need not align with disjoint neuron sets (Geiger et al., 2024). The comparison point for controlled algorithmic tasks is grokking and progress-measure work (Power et al., 2022; Nanda et al., 2023). Those papers study delayed generalization and clean learned algorithms; here the aim is a formation audit for a retrieval role whose write side is less closed-form.

Prior work also shows that circuit algorithms can remain stable while component identities change. Tigges et al. track circuits across training and scale and find that the algorithm can persist while the implementing heads vary (Tigges et al., 2024). I therefore do not claim that role/address dissociation is new by itself. The contribution here is the controlled from-initialization audit: behavior, route geometry, causal tests, optimizer-update attribution, and cross-seed address movement in one small model.

The optimizer accounting is specific to AdamW, building on Adam (Kingma & Ba, 2014) and decoupled weight decay (Loshchilov & Hutter, 2017). The write-side intervention style is related to causal tracing and model editing (Meng et al., 2022); the gradient-based attribution logic is also adjacent to faithfulness-oriented circuit attribution (Hanna et al., 2024). Mechanistic data attribution traces interpretable units to influential training examples in larger LLMs (Chen et al., 2026); my experiment instead traces a controlled role scalar through actual checkpoint updates. The novelty is not QK/OV decomposition, progress measures, causal intervention, or cross-seed comparison individually. It is their combination in a single formation audit.

3. Task and Model

The task is symbolic latest-write key-value lookup. A sequence contains write events and read events:

```
W K03 V14   W K01 V09   R K03
      W K03 V02   R K03.
```

The correct answer for the final read is `V02`, not `V14`, because `V02` is the latest previous write for key `K03`.

Although the task is synthetic, it is not a one-step lookup table. It requires three separable internal operations: identify the queried key, retrieve the latest matching support value rather than a distractor or stale value, and write value identity into the prediction-position residual stream.

The main model is a 3-layer decoder-only transformer with 4 heads per layer, $d_{\text{model}} = 128$, $d_{\text{ff}} = 512$, no dropout, and 626,048 parameters. It is trained with AdamW (Kingma & Ba, 2014; Loshchilov & Hutter, 2017), learning rate 4×10^{-4}, betas $(0.9, 0.95)$, weight decay 0.01, gradient clip 1.0, batch size 128, 16,000 training steps, and 200 warmup steps. I use two related seed-7 runs: a sparse-checkpoint heldout-generalization run for model selection, and a dense-checkpoint formation run for SVD, causal patching, and exact optimizer accounting. The sparse run selects performant reference behavior; the dense run is used for formation accounting because it saves optimizer state and frequent checkpoints. I do not treat the dense run as an independent cross-seed replication. Most exact formation claims use the dense formation run over the $0 \to 6000$ optimizer-trace horizon.

The benchmark includes shortcut checks. Exact sequence

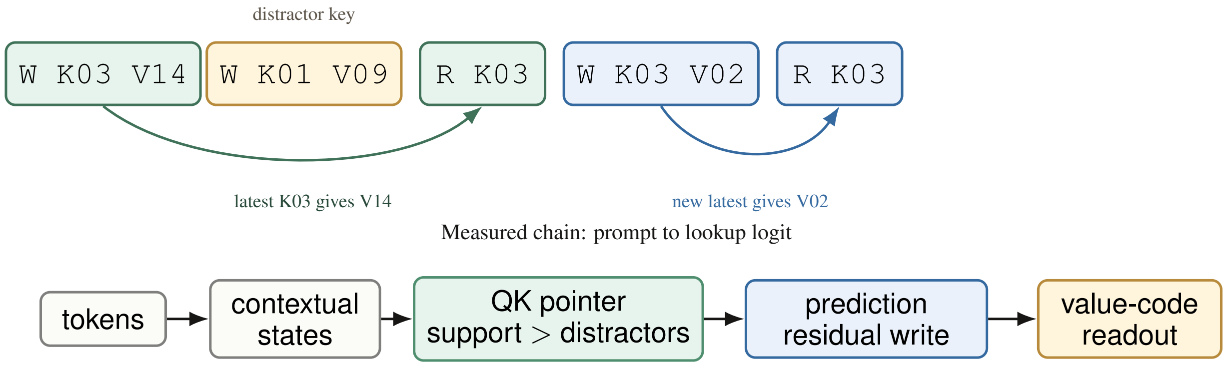

Figure 1. Task and evidence chain. The model must return the most recent value written for the queried key, and I audit the learned computation from tokens through contextual states, QK routing, residual writing, and value-code readout.

overlap across splits is zero, latent-program overlap is zero, and heldout leakage outside the heldout split is zero. Trivial heuristics are weak: the strongest tested frequent-value heuristic reaches only about 0.146.

4. From Components to Roles

The analysis did not begin with the QK route. I first tried component maps, feature-family analyses, and neuron interventions. These found real load-bearing structure, but not clean atoms of computation. Early components were often causally important even when their direct logit attribution was weak or pointed the wrong way. Feature-family analysis exposed superposition (Elhage et al., 2022; Bricken et al., 2023): candidate stories shared neurons, opposed each other through other neurons, and contained sign-conflicted units. Neuron interventions did not isolate a clean variable.

One failure was especially useful. A feature-family birth model predicted the wrong family would form first, even though the eventual winner was more useful for heldout behavior. The failure suggested that feature-level gradient proxies were insufficient; I needed to track the role being written. This is the point where the analysis moves from component names to role-level scalars.

I therefore define role-level objects. The route object is:

$$C_{QK}(\theta) = \mathbb{E}_x[\Delta s_\theta(x)],$$
$$\Delta s_\theta(x) = s_\theta(p_x, v_x) - |D_x|^{-1} \sum_{d \in D_x} s_\theta(p_x, d),$$

where p_x is the prediction position, v_x is the true support-value position, and D_x is the set of value distractor positions in example x. In words, C_{QK} asks whether the prediction position scores the true support value above distractor values.

The write object is:

$$C_{\text{write}}(\theta) = \mathbb{E}_x[\, g_{\text{ref}}(x)^\top \Delta h_{\text{write},\theta}(x)\,].$$

In words, remove a source, measure the missing residual change, and ask whether that missing change points in a mature answer-relevant direction g_{ref}.

5. QK Route Birth

The QK side is the most complete part of the account. I use the standard transformer-circuits decomposition where QK terms route attention and OV/write terms move residual information (Elhage et al., 2021). In the reference seed, a route through $L2H1$ becomes a low-rank support-value matcher. The relevant matrix is:

$$W_{QK}^{(\ell,h)} = W_Q^{(\ell,h)} W_K^{(\ell,h)\top}.$$

Tracking this matrix over checkpoints shows singular-value concentration, effective-rank collapse, and stabilization of top singular directions during the formation window.

The route is not merely a post-hoc attention pattern. First-order route attribution asks whether actual checkpoint updates predict route growth:

$$\Delta C_{QK} \approx \nabla_\theta C_{QK}(\theta_t)^\top \Delta\theta_t.$$

Using the actual AdamW parameter update from the traced run, this approximation tracks the realized route movement. In the from-initialization trace, actual route growth is $+4.11462$ and the AdamW reconstruction predicts $+5.21734$. The raw-gradient, SGD-equivalent term accounts for only about 0.76% of the actual growth.

The word "exact" here applies to the parameter update, not to the scalar change. The measured $\Delta\theta_t$ is the actual AdamW update between checkpoints; the scalar attribution is still first-order.

A causal patching check, in the style of prior causal circuit-localization work (Goldowsky-Dill et al., 2023; Conmy et al., 2023), supports this route interpretation. At the final checkpoint, full `layer_1_post_mlp` residual patching recovers the clean-corrupt margin difference by 40.58,

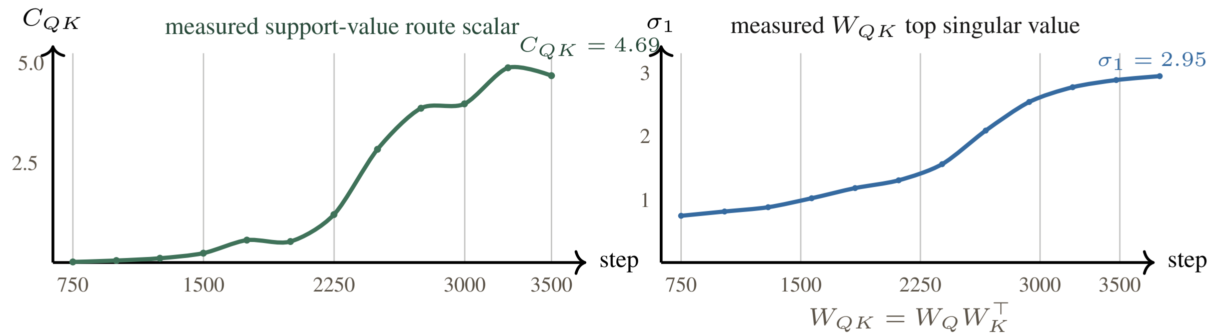

Figure 2. QK birth. The reference route becomes visible in measured C_{QK} growth, measured W_{QK} singular-value growth, and support-value-over-distractor separation.

while the rank-4 $L2H1$ QK query route transfers 10.54, or 25.98% of the full residual transfer. The same route has near-zero distractor transfer (-0.317), suggesting it is aligned with the query-key/support relation rather than arbitrary prompt structure. This also shows the mechanism remains distributed: the best small route is important, but not the whole circuit.

The update is also not symmetric across the two sides of W_{QK}. In a traced 5500–5550 window, the key-side term for the leading route actively opposes growth, while the query-side term carries nearly all of the improvement. This matters because a QK matcher is a bilinear object: route formation can happen by moving the query side, the key side, or both. In this case, the useful movement is concentrated on the query side.

6. Optimizer Accounting

The raw gradient is not the wrong object, but it answers a different local-slope question. During the dense competition phase, many candidate routes are active, and their instantaneous gradient contributions can cancel. AdamW's preconditioned update is a different object: it rescales directions using historical second-moment information and, when $\beta_1 > 0$, also carries first-moment state.

The matched seed-7 optimizer ablation tests the boundary of this claim. Under the tested seed-7 recipe and budget, AdamW variants learn the task and form the route, while tested SGD variants do not. AdamW baseline reaches validation answer accuracy 0.976 with QK separation 8.03. AdamW with $\beta_1 = 0$ reaches 0.984 with QK separation 9.26. The best SGD+momentum learning-rate sweep run reaches only 0.0085 validation answer accuracy, and the best observed SGD QK separation is about 0.118.

The $\beta_1 = 0$ result is important. It means the strongest statement is not simply "momentum did it." In this setup, first-moment momentum is not necessary for the route to form. The sharper hypothesis is that adaptive preconditioning, especially second-moment scaling, makes the route basin accessible under this budget.

Table 1. Matched seed-7 optimizer ablation. This is not a universal proof that SGD cannot learn the task; it is evidence that AdamW-style adaptive preconditioning is efficient under the tested budget and recipe.

variant	ans. acc.	QK sep.
AdamW baseline	0.976	8.03
AdamW $\beta_1 = 0$	0.984	9.26
best SGD+momentum sweep	0.0085	≈ 0.118

7. Role/Address Dissociation Across Seeds

The role-level interpretation predicts that the same computation may appear at different component addresses. Figure 4 summarizes the cross-seed scan across five additional seeds.

For each seed, I ranked heads by the same predefined support-value route scalar C_{QK} on the same fixed probe set. Winner heads are the top positive movers; controls are bottom-ranked or weak/negative movers under the same scalar. The winner was selected by this scalar, not by manual inspection of the final head map.

Winner heads grow positively in the support-value route direction, while bottom controls move negatively or weakly. The cross-seed result changes the interpretation from "head $L2H1$ is the circuit" to "a support-value retrieval role is repeatedly instantiated, and the address is seed-dependent." This complements prior evidence that circuit algorithms can remain stable across training and scale even when head identities vary (Tigges et al., 2024); here the moving address is measured in a controlled formation audit.

This is the main conceptual interpretation of the cross-seed scan: in this setup, the measured role is more stable than the component address.

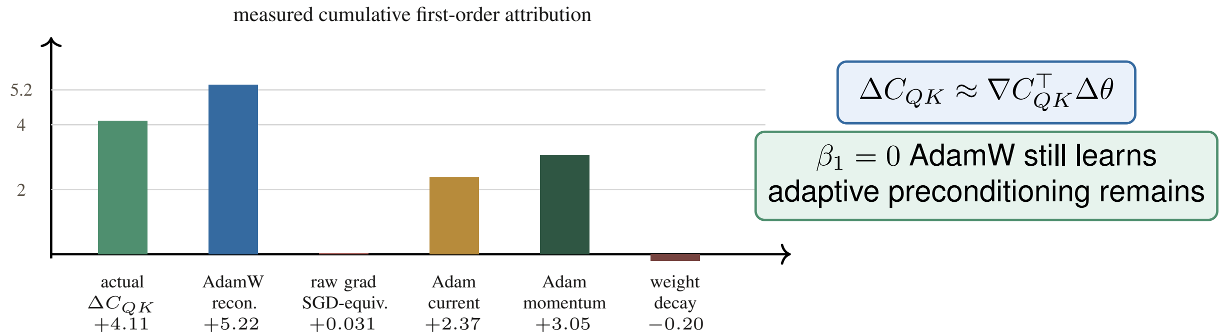

Figure 3. Optimizer accounting. The measured route growth is tracked by first-order attribution using the actual AdamW parameter update. The raw-gradient, SGD-equivalent direction is tiny in the traced reference run.

same role, different address; winners grow, bottom controls do not

seed	QK winner	winner ΔC_{QK}	bottom	bottom ΔC_{QK}	write/readout winner
0011	L2H0	+1.448	L0H0	-0.190	L1H3→L1MLP
0013	L2H2	+1.451	L1H1	-0.230	L1H3→L1MLP
0017	L2H3	+3.178	L0H2	-0.254	L1H1→L1MLP
0023	L2H1	+1.500	L1H2	-0.114	L2H1→L2MLP
0029	L1H2	+1.439	L1H0	-2.577	L1H1→L1MLP

role repeats
address moves

Figure 4. Role/address dissociation. The support-value retrieval role repeats across seeds, but the winning head address changes. The quantitative columns report actual route growth over the 750–2500 attribution window for the selected winner and the bottom control.

8. The Write Side Is Contextual, Not Static OV

The write side does not mirror the QK story. I do not find a clean static W_{OV} theorem analogous to the low-rank W_{QK} route. The useful object is instead a contextual residual perturbation at the prediction position.

At `layer_2_post_mlp / prediction`, removing the rank-16 value-identity subspace drops validation margin by 4.97 and validation answer accuracy by 0.288 at step 3500. Keeping only a low-rank value subspace fails, but keeping rank 127 value identity almost preserves IID behavior: validation accuracy changes from 0.7647 to 0.7582. The rank-127 result should not be interpreted as a compact-code result. In a 128-dimensional residual stream, preserving 127 dimensions is close to preserving the full state. I read this as evidence against a small low-rank value-vector account, and in favor of a broadly distributed value-readable prediction-position state. The value code is broad, but it is causally used.

The transfer tests clarify the computation. A source-only map fits:

$$\hat{z}_{\text{prediction value}} = A z_{\text{support value}} + b.$$

This helps, but does not close the stable write/readout scalars. A contextual map fits:

$$\hat{z}_{\text{prediction value}} = A z_{\text{support value}} + B z_{\text{prediction context}} + b.$$

This source-plus-context transfer nearly restores stable write/readout scalars: at step 3500, it rescues 1.005 of negative-answer-loss damage and 0.754 of fixed removed-branch margin damage. However, context-only rescue is already high (0.959 and 0.640 on the same two scalars). The support value does not write the answer code from scratch; it helps shape a prediction-position scaffold that is already partly readout-ready.

The current write-side claim is therefore empirical and bounded. I have causal evidence for a broad prediction-position value code, transfer evidence for source-plus-context reconstruction, and optimizer evidence for write/readout growth. I do not yet derive a closed-form construction of the prediction-position scaffold.

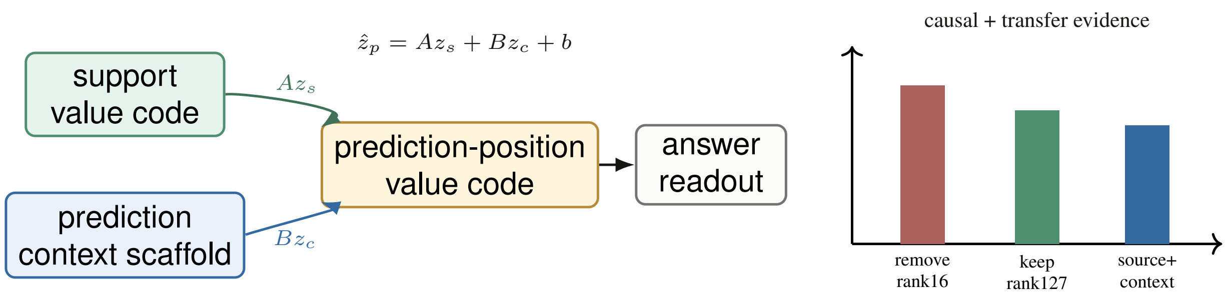

Figure 5. Write/readout side. The write side is best characterized as a contextual prediction-position value-code operation, not as a single static OV vector.

9. A Methodological Trap: Moving Margins

The usual answer margin can hide a real circuit. The correct logit is fixed by the task, but the "best wrong token" can change across checkpoints or along an interpolation path. When this branch changes, the moving margin

$$\mathrm{logit}(\mathrm{correct}) - \max_{w \neq \mathrm{correct}} \mathrm{logit}(w)$$

is not one smooth scalar. It is a piecewise object, so endpoint-gradient and line-integral diagnostics can fail even when the internal route is real.

Fixed-branch and output-space scalars are safer formation targets. In the 1500–2500 window, route/write scalar closure explains correct-value logit with about $R^2 = 0.37$, while output-space closure reaches 0.868 in the matched fixed-scalar run. Moving answer margin remains harder, reaching about 0.407.

10. Limitations

This is a controlled mechanistic case study, not a universal theory of transformers. The main limitations are:

- The task is synthetic and small. This is useful for auditability, but it does not establish scaling.
- The optimizer ablation is bounded. I do not claim plain SGD can never learn the task; I claim the tested same-budget SGD variants did not.
- The write side is not fully closed. I have a contextual value-code account, not a closed-form derivation of the prediction-position scaffold.
- Full moving answer-margin closure remains partial. Fixed-branch and output-space scalars are cleaner formation targets.
- Neuron-level decomposition remains blocked by superposition. Subspace-level claims are more faithful to the evidence.
- The role/address result uses six total seeds in one task family. It supports role recurrence in this setup, not a general theorem about transformer circuits.

11. Conclusion

In this controlled transformer, the measured role is more stable than the component address. In the traced reference run, the QK side of the lookup role becomes visible as a low-rank support-value matcher, and first-order attribution using the actual AdamW parameter update tracks its growth better than the instantaneous raw-gradient direction. Across seeds, the role repeats while the winning head changes. The write side is real but different: a contextual, high-rank value-code operation at the prediction position rather than a clean static W_{OV} theorem. The result is a reproducible formation audit: strong for QK, supported and causal for write/readout subspaces, and explicit about what remains open.

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A. Reproducibility and Artifacts

The submission is anonymized for double-blind review. The reproducibility package should contain the training configs, figure-generation script, CLI entry points, and compact analysis artifacts needed to reproduce the main tables and figures. The main artifact families are: QK route birth, optimizer-update attribution, cross-seed head selection, write-side functional subspaces, value-code interventions, contextual transfer rescue, and closure diagnostics. Large raw checkpoint sweeps are not needed for review if the compact derived artifacts and commands are included.

Each main claim in the paper has a corresponding artifact family: C_{QK} growth and W_{QK} SVD rows for QK birth; AdamW decomposition rows for optimizer accounting; fixed-probe cross-seed scans for role/address dissociation; residual intervention and transfer-rescue rows for the write/readout claim; and branch-aware/fixed-output scalar rows for the moving-margin limitation.